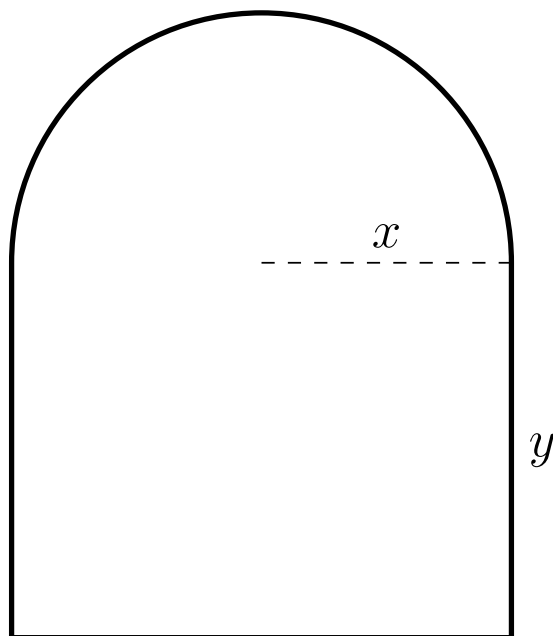


Problem 2

Problem 2

Problem 1 A mirror has the shape of a rectangle surmounted by a semicircle (see figure). The area of the mirror is 32 in^2 . Let x be the radius of the semicircle (that lies on top of the rectangle). Express the perimeter of the mirror P (in inches) as a function of x (in inches). Is P a polynomial, a rational function or a transcendental function?



Explanation. Since the area $A = 32 = 2xy + \frac{x^2\pi}{2}$, it follows that $y = \frac{16}{x} - \frac{x\pi}{4}$ and that $P(x) = 2x + 2y + x\pi = 2x + \frac{32}{x} - \frac{x\pi}{2} + x\pi = 2x + \frac{x\pi}{2} + \frac{32}{x}$